

Commands

Creating a file

```
echo "Name: Abdul Gafoor Mubashir, Netid: Vq4891, I am Feeling Happy " >> plaintext.txt
```

Change Permissions on this file

```
chmod 600 plaintext.txt
```

Symmetric encryption with binary output

```
gpg --symmetric plaintext.txt
```

#Changing name of encrypted file

```
mv plaintext.txt.gpg ciphertext.txt.gpg
```

Revealing Ciphertext

```
cat ciphertext.txt.gpg
```

Symmetric encryption with ascii output

```
gpg --symmetric -a plaintext.txt
```

#Revealing Symmetric encryption with ascii output

```
cat plaintext.txt.asc
```

Setup some prearranged listener

```
nc -lvp 1234 > ciphertext.txt.asc
```

lookup ip of listener

```
ip addr
```

#Sending our ascii armored, encrypted file to some prearranged listener

```
cat plaintext.txt.asc | nc 10.0.0.99 1234
```

Decrypt the asymmetric message

```
gpg --decrypt ciphertext.txt.asc
```

Import asymmetric public key

```
gpg --import csmith.pub.key
```

List the imported keys in local keyring

```
gpg --list-keys
```

Create a public/private key pair

```
gpg --gen-key
```

#List Keys

```
gpg --list-keys
```

Export the public key in ascii format

```
gpg --export -a > VQ4891.public.key
```

Sign the plaintext file with my private key

```
gpg -a --output plaintext.txt.asc.sig --sign plaintext.txt
```

Encrypt the signed file to the recipient's key

```
gpg -e -a -u "20AD3F6C8F0A8F65B733D9A2147A464DDE1FBB2A" -r  
"17EC6F82BA35580BA1C9613EB515325350F68884" plaintext.txt.asc.sig
```

install stego tools

```
sudo apt-get install steghide
```

#checking manual of steghide

```
man steghide
```

Download an image file

```
wget https://i.pinimg.com/originals/be/96/ce/be96ce958f2e678cb57b38c26cecac42.jpg
```

#change name of image downloaded

```
Mv be96ce958f2e678cb57b38c26cecac42.jpg car1.jpg
```

Embed a plaintext message in an image file

```
steghide embed -cf car1.jpg -ef plaintext.txt -sf steg_image.jpg
```

Check integrity

```
For Original image: md5sum car1.jpg
```

```
For embedded image: md5sum steg_image.jpg
```

```
For renamed original image: md5sum renamed.jpg
```

Extract your secrets

```
steghide extract -sf steg_image.jpg
```